

MAT7058 Combinatorics II, Exercises 1

Due Date: 04 March 2025, Wednesday, beginning of class.
Only handwritten paper submissions in English are accepted.

- Each exercise is worth 4 points.
 - In general, you have to prove everything and only use things which were introduced in class. Ask if you are unsure.
- 1 Let A be the adjacency matrix of a graph $G = (V, E)$.
 - (a) Find a combinatorial interpretation for each entry of A^2 and A^3 .
 - (b) Show that $\text{tr}(A^2) = 2|E|$.
 - (c) For $m \geq 2$ an integer, what does $\text{tr}(A^m)$ count? Prove your claim.
 - 2 The *girth* of a graph G is the length of the smallest polygon in the graph. Let G be a graph of girth 5 for which all vertices have degree at least d . Show that G has at least $d^2 + 1$ vertices. Can equality hold?
 - 3 Let $Q = \{1, \dots, q\}$ be the graph with vertex set Q^n , two vertices adjacent if they differ in exactly one coordinate. Show that Q is Hamiltonian.
 - 4 Find the six nonisomorphic trees on 6 vertices, and for each compute the number of distinct spanning trees in K_6 isomorphic to it.
 - 5
 - (a) Suppose a tree G has exactly one vertex of degree i for $2 \leq i \leq m$ and all other vertices have degree 1. How many vertices does G have?
 - (b) Suppose a tree G has exactly one vertex of degree i for $2 \leq i \leq m$ and k other vertices have degree 1. Prove that $k \geq \lfloor \frac{m+3}{2} \rfloor$ and give a construction for such a graph.